■ Soil Health & Crop Recommendation Report

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Farmer Information	
Name:	bapre
Phone:	2372352323
Location:	tonsyal
Land Size:	100.00 acres

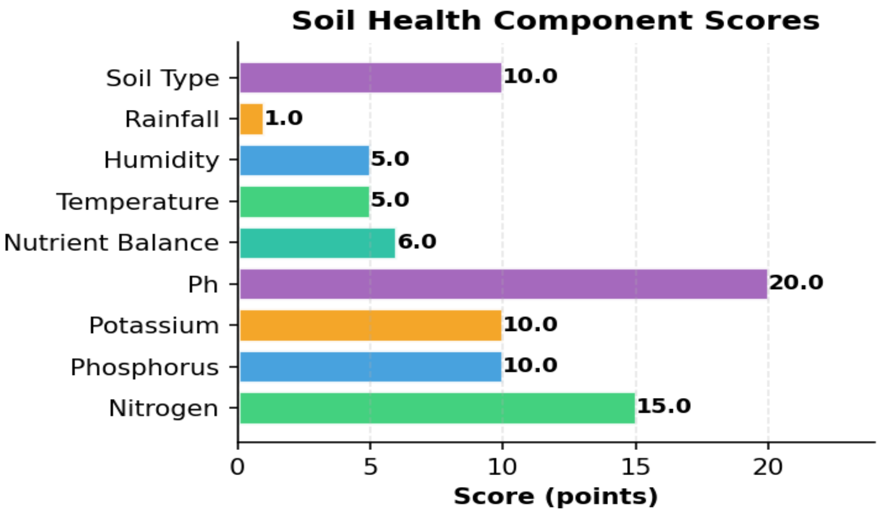
Soil & Weather Data

Parameter	Value	Unit
Nitrogen (N)	61.0	mg/kg
Phosphorus (P)	70.0	mg/kg
Potassium (K)	81.0	mg/kg
pH Level	6.50	
Temperature	28.9	°C
Humidity	72.8	%
Rainfall	12.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score [illegible]	.180392,.8,.44313 [illegible]	[illegible] 82.0/100 [illegible]
Nitrogen	15.0	points
Phosphorus	10.0	points
Potassium	10.0	points
Ph	20.0	points
Nutrient Balance	6.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	1.0	points

Soil Type	10.0	points



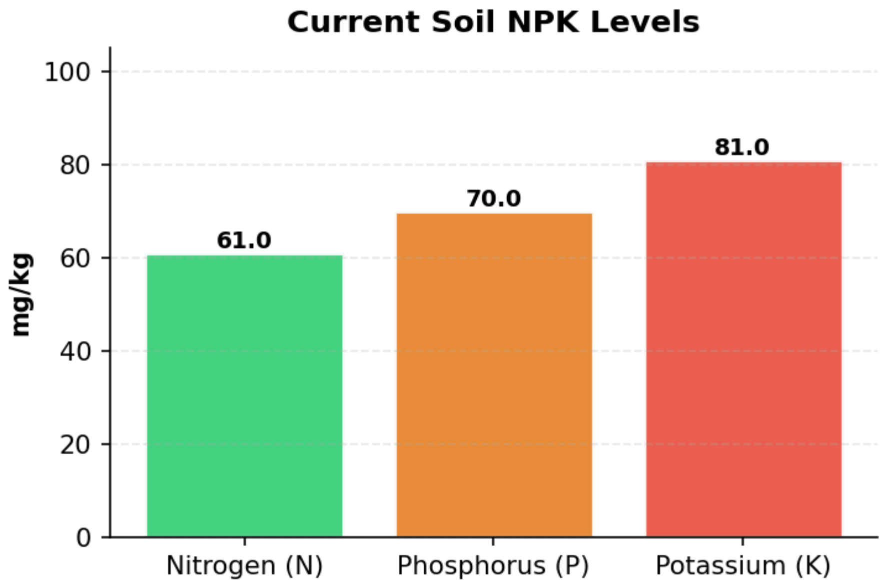
Recommended Crop

Predicted Crop	Muskmelon
Model Used	Stacking Classifier
Confidence Level	62.1%

Alternative Crop Options

Rank	Crop	Confidence
2	Banana	11.1%
3	Blackgram	1.9%

Soil NPK Levels Visualization



Soil Treatment Recommendations

• Soil pH (6.5) is within optimal range (6.0-7.0) for muskmelon.

• Nitrogen (61.0 mg/kg) is within optimal range for muskmelon.

• Phosphorus (70.0 mg/kg) is excessive for muskmelon. Optimal range: 30-50. High P can interfere with micronutrient uptake.

• Potassium (81.0 mg/kg) is excessive for muskmelon. Optimal range: 50-70. High K can cause magnesium deficiency.

• Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

■ **Fertilizer Requirements for muskmelon:**

• Land Size: 100.0 acres

• Current NPK: N=61.0, P=70.0, K=81.0 mg/kg

• Ideal NPK: N=80, P=40, K=60 kg/acre

Required Application:

• Nitrogen (N): 1900.0 kg total (19.0 kg/acre)

→ Apply in 2-3 split doses: 50% at sowing, 25% at tillering, 25% at flowering

■ **Organic Alternatives:**

• Compost/FYM: 5-10 tons/acre (provides N, P, K)

• Green Manure: Leguminous crops (fixes N)

• Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Muskmelon

Requires warm climate and well-drained soil. Provide support for vines.

Additional Notes

• These recommendations are based on comprehensive soil analysis and advanced ML predictions.
• Actual results may vary based on local conditions, weather patterns, and farming practices.
• Consult with local agricultural experts before making major farming decisions.
• Regular soil testing (every 2-3 years) is recommended for optimal crop management.
• Consider crop rotation and organic farming practices for long-term soil health.
• Monitor weather forecasts and adjust irrigation schedules accordingly.